Things, places, people & stories - the platform

Designing a platform for genre experiments with locative media in the cultural heritage sector

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Abstract

While working with prototypes for digital genres for the cultural heritage sector, I found a need for a technology platform that makes it easier to start experimenting with prototyping and genre development for the sector. The design of such a platform then became a sub-project within my PhD-project. The design is based on requirements gathered from the research, the main project, and other related projects. The platform is implemented as a Web API, using REST as architectural style, and Atom Publishing Protocol as basis, but using extensions through a plug-in architecture. It is now being tested through prototyping in different sub-projects.

Introduction

This paper describes a practical sub-project within my PhD project, *Things, places, people & stories* (TPPS), which again is part of a larger research project, Inventio. The objective of TPPS and Inventio is to research genre development in digital media with the use of mobile and location based technologies. The research combines practical prototype development with theoretical issues in a humanities perspective, and is largely done in collaboration with museums and cultural heritage institutions.

Such an approach opens up a whole range of practical issues related to development. In the Inventio-project, we see that there is a great deal of overlap between the different prototype's requirements that we have failed to address effectively. We also see the same overlap when analyzing different projects and services within the cultural heritage sector. We therefore see a need for a platform for easier development and experimentation with genres in connection with the museums' and other cultural heritage institutions use of digital media. This paper describes the design of such a platform within the scope of the overall project's objectives.

Background

The requirements for this platform come mainly from three sources:

- The project's research with its theories, methods and models.
- Expectations regarding the use of digital and social media.
- Existing genres within the institutions.

These requirements are more in the form of constraints, suggestions and guidelines, than formal requirements in the sense usually found in commercial software development projects.

Theory, methods and models

The Inventio project, of which TPPS is a part, is theoretically grounded in new rhetoric and genre theory, in combination with design theory, computer science, activity theory and socio-cultural learning theory. In TPPS I use a triangular model (Fig. 1) to illustrate the complexity of the field.

Axis A), the *genre axis*, illustrates the relationship between the (rhetorical) situation (Bitzer, 1992) and text production, based on Carolyn Miller's understanding of genre as "typified rhetorical actions based in recurrent situations" (Miller, 1984). Text is here understood as multimodal composite expressions using different types of semiotic resources (Kress and Van Leuwen, 2001).

Axis C), the *technology axis*, describes the relationship between the situation, understood both as a historically concrete situation (*this* museum *now*) and as socially defined context (museum as *institution*), and technology development.

Axis B), the *media axis*, shows the dependency between technology and text production in digital media.

The triangle is a reminder that the axes and corners are not independent, and must be analyzed in context. In the light of this model, we can say that the practical development of the platform takes place along the *technology axis*, to enable experimentation along the *media axis*, based on analysis of the *genre axis*. The goal of the experimentation is to produce forms of digital multimodal texts that can function as genre prototypes. The binding element in this model is design. That is not the only option, but one way to elaborate the intricacy of the relations and interactions involved.

The relationship between genre and design can be described in terms of the relationship between *typified response* and *design pattern*. *Design pattern* is a term originally used in connection with architecture, where it describes common solutions to common design problems (Alexander et. al, 1977), but is now an established concept in software design. This relationship between genre and design is modeled on Miller's placement of genre in a hierarchy of aggregates of form and substance (Fig. 2). Form and substance at one level constitutes the substance of the form on the level above, and the levels are linked together by actions. In this model, *genre as action*, applies a *design pattern* as a *typified response*, in response to recurrent rhetorical situations. Since genre-mixing is seen as an important source of genre development (Liestøl, 2006; Fairclough, 2003; Asdal et al. 2008), we let this model be recursive to cover aggregations of design patterns and typified responses (Fig. 3). Again, this is not the only way to perceive genre, but it lets us work with genre in regard to form and composition in a novel way that is suited for development.

Some concepts of genre-mixing are important for a design of a platform for genre experiments:

- *Genre Convergence* describes genre mixing using a convergence model (Liestøl, 2006).
- *Format* (Fairclough, 2003) describes how different genre elements and expressions may be part of a larger multimodal text unit.
- *Genre Chain* (Swale, 1990) is a concept describing how different genre expressions form chains within a given context.

These are all useful concepts when working with genres in digital media. The *Convergence* model can, for example, be used to describe the relationship between podcasts and general news syndication. *Format*, Fairclough already uses in connection with websites that combines genre elements (I will use the term genre format to distinguish it from the more common data format). *Genre chain* is a very useful concept for describing forms of asynchronous communications, such as the relationship between an article and a comment.

The media axis can be described as a systemic relationship between technology and text. Such a system is often described using a layered model (Liestøl, 2006, Platformstudies.com, 2010, Lüders et al., 2010). Liestøl uses a tripartite model consisting of hardware, software, and what he calls meaningware (Fig. 5), which is the level of text and genres. The relationships between the various levels are intricate, but can be described by means of design. Each level defines the *design space* for the level above. This design space is defined negatively by the *constraints* set by the underlying level, and positively by the *possibility space* the underlying level opens up. We can compare it to chess: board, pieces and rules are constraints, while the possibility space is equal to the set of possible chess games. Design pattern can in this analogy be compared to strategies and patterns for playing, like the opening game called Sicilian Defense.

Expectations regarding the use of digital and social media in the museum and cultural heritage sector today

A part of the C)-axis is the different expectations we have for the museums to use new digital media. These expectations must be seen in the context of the traditional and general responsibilities for education, documentation and public exhibitions and services, and they are linked to a number of factors and trends that affect the museums rhetorical situation. The key trends for Inventio and TPPS are:

- increased use of mobile devices,
- competition with more immersive media, like video games, for the attention of the young
- general trends on the Web, covered by the concept of Web 2.0
- a shift towards social media like Facebook and Twitter
- increased focus on location-based services, such as Four Square and Gowalla

Some of these trends are summed up in the recent *2010 Horizon Report: Museum Edition* (Johnsen et al., 2010).

Expectations emerging from trends like these, expresses some overall guidelines for the design of this platform. The most complicated of these, is the competition with immersive media. Within the Inventio-project, we can see *Situated Simulations*, with the use of Augmented Reality and 3D, as a kind of response to this expectation (Liestøl, 2010; Johnsen et al., 2010 p. 16). It is limited what we can build of support for it directly into the platform, except for support for the Layar AR-platform (Layar, 2010) that is included as alternative presentation format. But immersion is also about narrativity, and that is within the scope of the platform.

Existing genres and expressions

It is important that the platform supports already popular digital genres in connection with museums and such. This means genres like digital guides, online exhibitions, and newer forms of interaction, like interactive walls in the exhibition space (Tallon et al., 2008; Warberg Løssing et al., 2009), all should somehow be supported.

Related projects

In addition we can find requirements for the platform documented through the projects in the Inventio-project, other museum initiatives, and various museum APIs.

Inventio project

The Inventio project consists of the following projects in addition to TPPS:

- *Situated Simulations* is a project working with dynamic reconstructions of historic buildings and events on site, using mobile technology, 3D design and Augmented Reality (Situated Simulations, 2010).
- *NarraHand* is a project about creating narrative structures in the city space using mobile phone technology and GPS, with an emphasis on collaborative storytelling (NarraHand, 2010).
- *Music in churches* is a project that uses mobile technology and GPS to combine tours of Rome to find the churches, with a guided tour within the churches, and using related classical music and composers to enhance the experience (Fagerjord, 2010).
- *Textopia* is a project for experimenting with location based literature and poetry, in a way that combines wiki or blog technology with mobile technology and the use of GPS (Textopia, 2010).

Related projects

There have been several projects related to museums and the cultural heritage sector that provide models for how such a platform can be designed. These projects provide a great opportunity to learn of what can be expected, and what works well, or not.

Museums APIs

Another trend that has contributed greatly to the design of this platform is the development of museum APIs (Application Programming Interface). It is a sector-specific variant of a more general trend about making own content accessible through Web-based APIs. Brooklyn Museum was an early adopter with its API (Brooklyn Museum, 2010), as well as the Victoria and Albert Museum (2010). These, together with others, like the Swedish project Swedish Open Cultural Heritage (SOCH, 2010) have made valuable contributions to understanding the needs for technology in the design of the platform.

Platform design

The platform is designed as a *Web API*. Unlike Web APIs designed as an add-on to, or as an interface to existing services, this platform is totally defined through its API. This means that the platform doesn't come with client applications, neither for management, publishing or otherwise. The reason for this is that it is precisely at the client level that we

want to start experimenting. It also gives us more flexibility in the choice of platforms for clients; desktop web, mobile web, native apps, or other handhelds.

The platform's main purpose is to make it easier to create clients, by clearly defining how they can communicate with an underlying service, and offering technology that makes it easier to set up such an underlying service. The platform provides functionality equivalent to a mini-CMS (Content Management System).

REST

The API is based on REST (REpresentational State Transfer), which is described as an architectural style for designing systems for distributed hypermedia (Fielding and Taylor, 2002). This design style is described through a set of design constraints. We can say that REST modifies the design space on a certain software level by introducing a set of soft constraints. REST has been chosen because it is more lightweight and flexible for experimenting with Web APIs than alternatives like SOAP and WDSL (Webber et al., 2010).

Atom Publishing Protocol

APP (Atom Publishing Protocol) is selected as the basis for the API, but is extended to provide a minimum level of CMS functionality. APP has been chosen for two reasons: 1) The Atom format is a simple generic hyper-text format for syndication, which can easily be extended, and 2), the protocol provides a good model for a service where you can also write back.

Atom format

An alternative to Atom format could be a more sector-specific format, based on sector standards, or semantic standards like RDF. One of the reasons for choosing Atom is because it provides an implicit fallback strategy. The Atom format can be read by services and systems that cannot make use of more sector-specific formats, but these formats can still be provided by using extensions for data that is more relevant for sector-specific contexts. This makes it easier to experiment with distribution and sharing, without losing the possibility for using rich semantic representations. Extensions are made by using the platforms plug-in architecture, and currently the platform supports the following extensions:

- GeoRSS for geodata.
- MediaRSS for distributing media.
- Rating for rating and voting functionality.
- Comments for commenting.

We are planning to integrate e-learning content as well, using extensions for e-learning standards.

Plug-in Architecture

If we take genre formats as a possible basis for genre mixing, in a form that require different types of data, there are specially two scenarios to work with them:

- As mashups where the API provides data that are combined with data from other services.
- And the convergence model, where the API provides ready-made combinations of different types of data, usually associated with different services, for the client to render.

Both of these scenarios are handled through the platform's plug-in architecture. If we want to combine a normal Atom entry with geodata, we use the GeoRSS-extension plug-in. By using this plug-in, the Atom entry functions as container for GeoRSS, and in that way can link an entry to a map. We can also use this plug-in architecture to extend the format with more sector-specific metadata. By using default Atom-elements for generic metadata, like title, and combining it with more specific metadata using the plug-in architecture, the platform provides the necessary fallback mechanisms by default.

The mashup scenario is supported by the Atom format on its own, but the plug-in architecture also provides functionality for establishing alternative service endpoints for other formats that might be needed in different mashup contexts. The plug-in architecture makes it relatively easy to add new formats and service endpoints in a modular fashion for specific experiments. Currently supported are:

- RSS (which also combines GeoRSS for mapping and media linking for podcasts),
- Layars JSON format for Augmented Reality navigation using smart phones (Layar, 2010),
- POI CSV for GPS devices
- oEmbed for sharing content between services (oEmbed, 2010).

The alternative endpoints provide us with the possibility to experiment with different technological platforms, and to enable a fallback strategy for services. If the scenario is an outdoor guiding of cultural heritage monuments, the tour could be made available as a high end AR-tour for advanced smart phones, through a custom mobile app that combines map data with rich media and smart navigation, by using Google Maps in general, as a podcast with the use of a MP3-player, or through a dedicated GPS unit or mobile with GPS software for consuming POI-data.

The implementation of the platform is based on PHP and MySQL. The installation requires minimum configuration, and login and registration is currently delegated to Facebook using their OAuth service (Facebook, 2010a).

Read / write service

The platform should make it possible to comment on published content, or rate in some fashion, or in any other way give feedback or participate in some sort of dialogue. This is crucial to meet the Web 2.0 and Social Media expectations concerning users and their contributions, dialog, and general participation.

In a genre perspective, the relationship between a blog entry and a comment is an example of a genre chain, and a lot of what we associate with Web 2.0 and Social Media can be analyzed as genre chains. That the API allows writing back, is a prerequisite for being able to experiment with genre chains, and Atom Publishing Protocol provides a very nice model for such functionality, since the format allows for developing more complex services and interactions through the use of the HATEOAS (Hypermedia as the Engine of Application State) principle, which is one of the often ignored constraints in REST based designs (Webber et al., 2010; Fielding, 2007). Atom is a genuine hypermedia format that offers rich, extensible, link functionality, which is central to HATEOS, and it means that

the extended format not only is suitable for data distribution, but also can be used for describing and defining a state based hypermedia protocol for data interaction (Webber et al., 2010). This includes creating advanced genre chains.

When working with genre experimenting in museums, we should both look at the existing genre chains (for example exhibition, educational services, classroom assessment, student report and teacher feedback), and how emerging digital genre chains can contribute to how museums solve their responsibilities and rhetorical situation(s).

It's there, But Is It ART?

As a test of the platform, we are now developing a prototype called *It's there, But is it ART?* The point of the service is to present art in the open with the combination of text, media, and geodata for mapping purposes, and then let the audience vote on whether the presented work really is art, or whether it is rather an expression of kitsch, design, advertising, or neither. You can also comment on the presentation or your vote. The service will be available via mobile (Fig. 6), and on web in a blog format (Fig. 7).

Genre-wise, the prototype will mix elements from blogs, social media, maps, guides, art reviews, rating systems, and polls. Both *convergence*, *genre format* and *genre chain* are useful concepts for describing the prototype. It mixes content similar to a blog entry with geodata, and voting is enabled through a special rating service created and configured for this specific purpose. This rating service is part of the platform.

Voting will require login via Facebook's OAuth service (Facebook, 2010a), and we are exploring the possibilities for further integration with Facebook (Fig. 8) using their Graph API (Facebook, 2010b). Using a single sign-on provider, like Facebook, will become a must for smaller services fighting for attention. By using standards, like OAuth and OpenID, enabled by social media services, like Facebook, Twitter, Google and many more, the threshold for registering with and later revisiting your service can be dramatically lowered.

Linking APIs together for distribution of a service, like we do when integrating the service with Facebook’s Graph API, shows how a genre chain on web can be extended by using

third party APIs. It shows the usefulness of the *genre chain* concept in addition to *convergence* and *mash up* when analyzing or creating combinations of data streams on the web. In addition, the use of different platforms like mobile, web and Facebook, shows how different platforms and contexts also have consequences for how the different genre elements are weighted in the mix.

Conclusion

The platform is still in its early stages and will be further developed and stabilized through several prototyping-projects. But we can already see in *It's there, But Is It Art!* that the platform will help us to move the focus and effort towards the interaction and media level of experimenting. The design space it offers, helps us both in practical development, and to reflect on theoretical issues, and thereby helping us to develop new genres for the cultural heritage sector.

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Figures

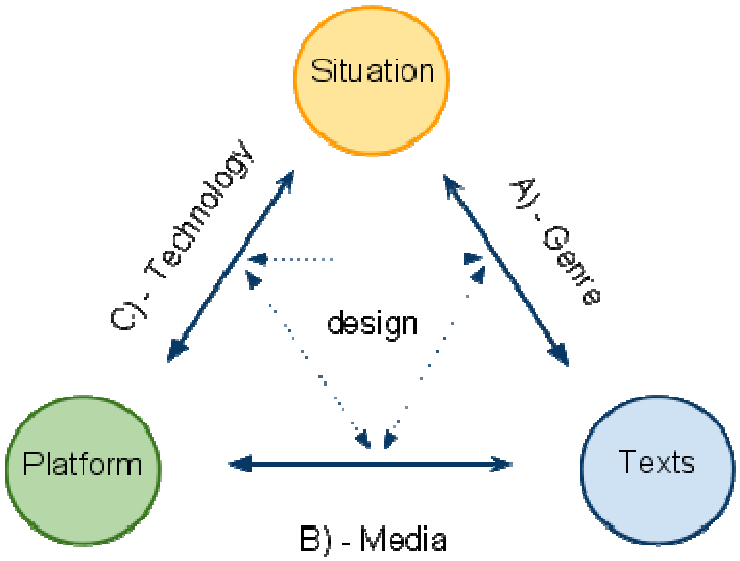


Fig. 1. Triangular model of the research field.

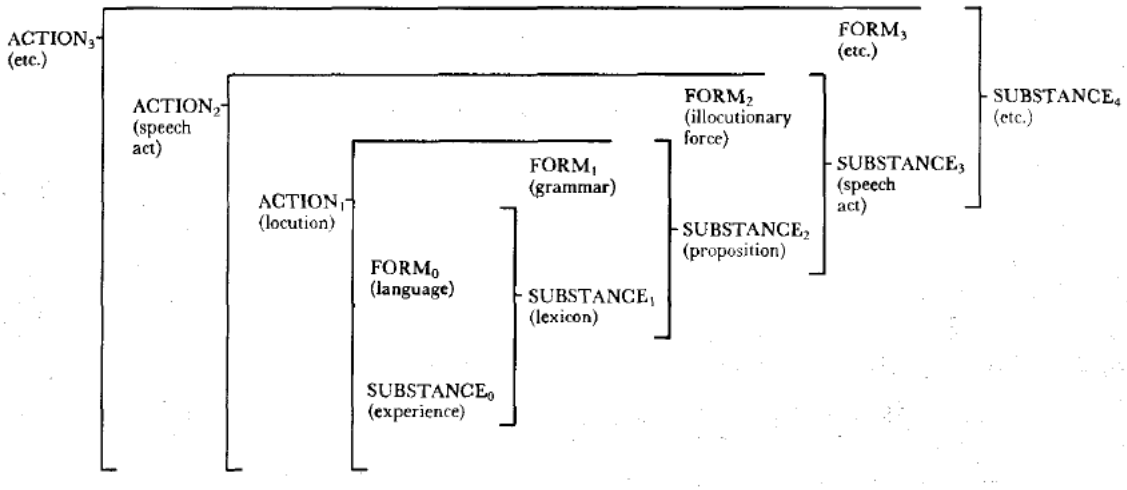


Fig. 2. “Hierarchical relationships of substance, form, and meaning-as-action. The combination of form and substance at one level becomes an action (has meaning) at a higher level when that combination itself acquires form. Each action is interpretable against the context provided by actions at higher levels.” (Miller, 1984, p. 160)

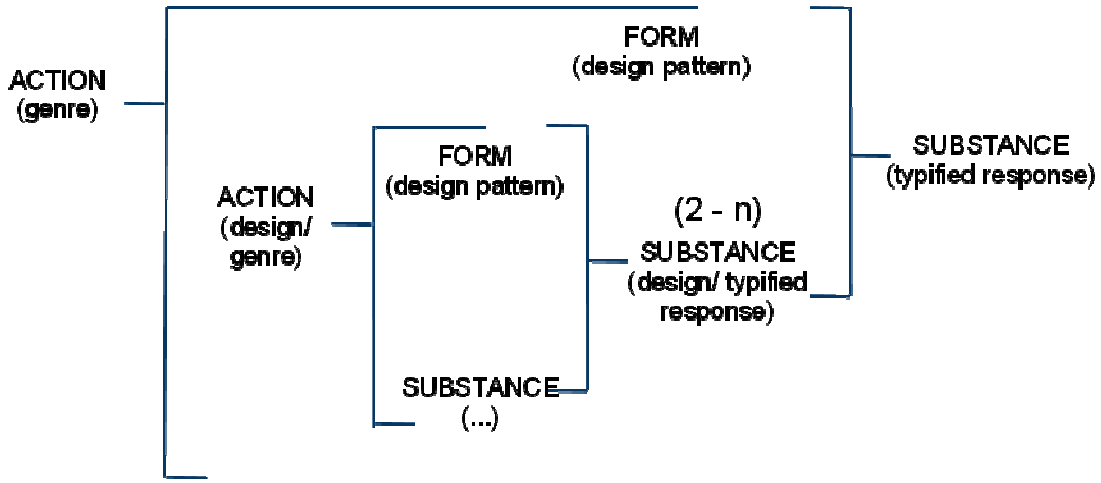


Fig. 3. Millers model applied on the (possible hierarchical/recursive) relationship between genre and design pattern.

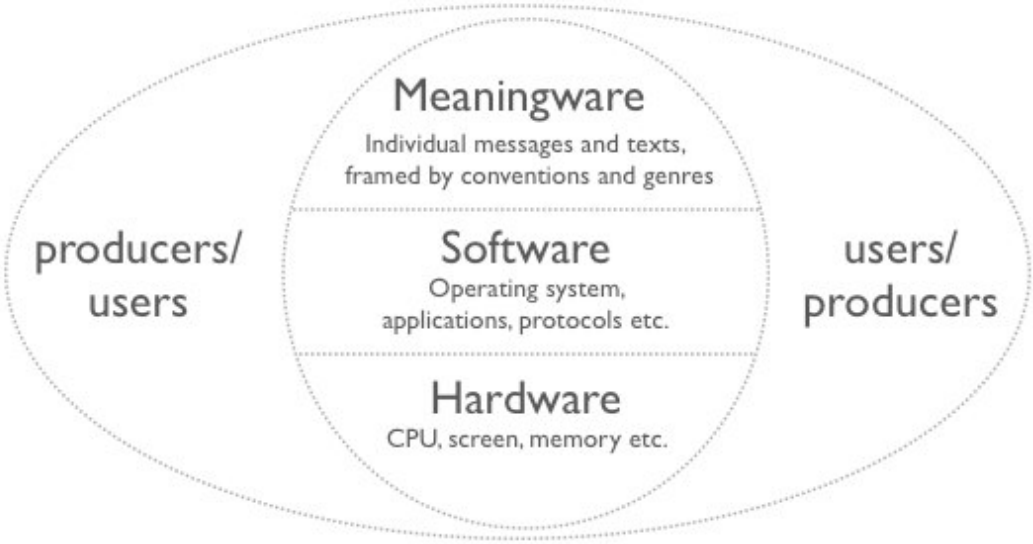


Fig. 5. Liestøls systemic model of digital media.



Fig. 6. A mock up of mobile client application for “It’s there, But is it ART?”

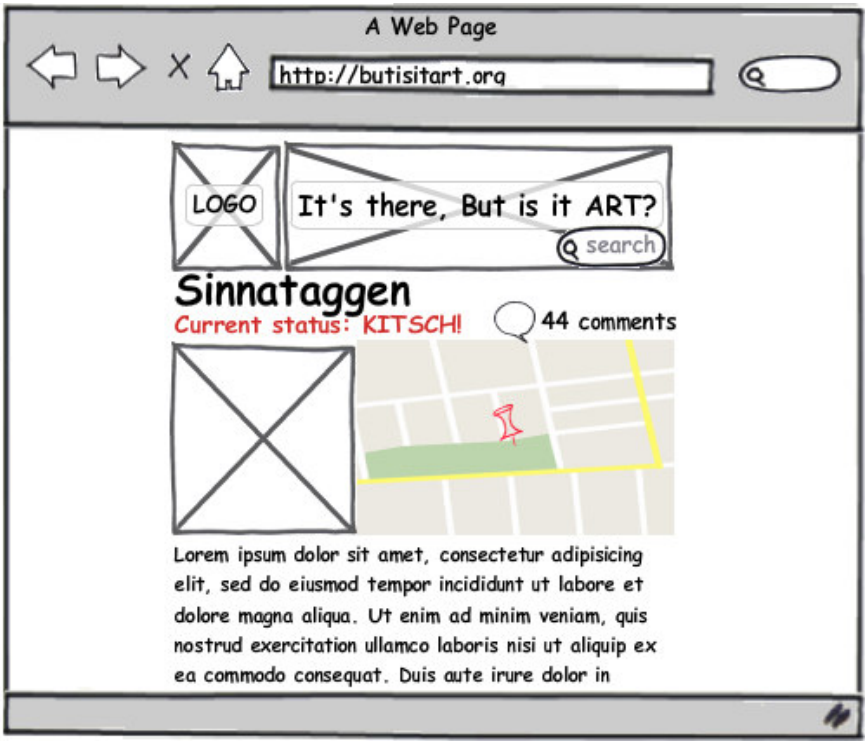


Fig. 7. A mock up of web application for “It’s there, But is it ART?”

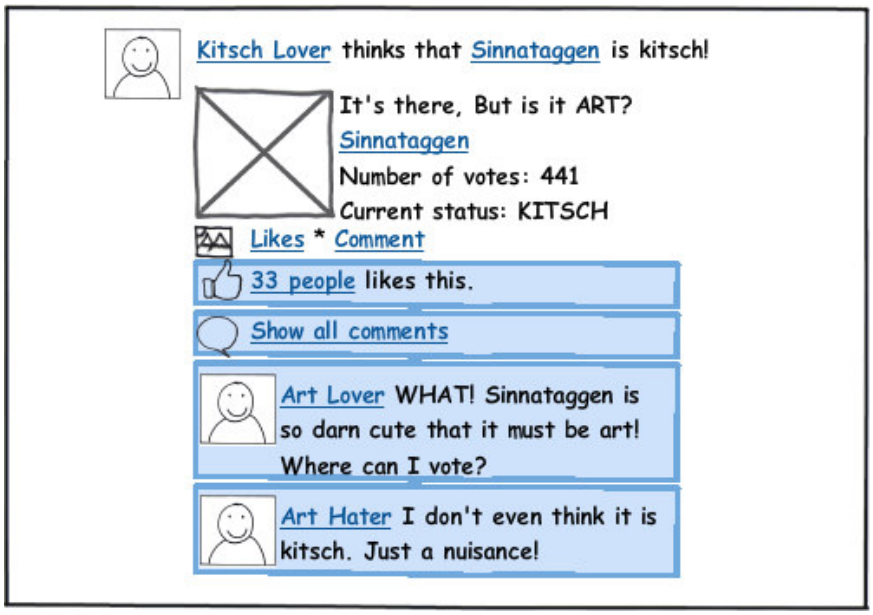


Fig. 8. A mock up of Facebook integration for “It’s there, But is it ART?”